

What AI Governance Needs to Survive the Intelligence Explosion

Proposal: Combine three things:

- Compute governance (track and report frontier training runs at the hardware level)
- Create a hybrid public / private certification model (market for certification providers / courts and certifiers handle deployment harms).
- US domestic regulation (agencies oversee, power stays spread across companies, export controls hold the China lead).

Component	Pros	Cons
Compute governance	Easy to monitor - a handful of chip fabs and data centers, not millions of users. Already happening (export controls, CHIPS Act).	Can't reach a model once it's already released or leaked. Compute is not a perfect stand-in for capability.
Public + private certification model	Uses courts that already exist. Creates a paper trail. Creates a marketplace for certification.	Risks conflict of interest that broke credit ratings in 2008. No company can ever pay enough to cover a true catastrophe, so liability doesn't work for the worst risks.
US domestic regulation	Not nationalized, but builds on what's already in motion AND includes public participation.	Lack of technical staff to actually check anyone's work. Without a crisis, enforcement stays weak.

Where it breaks:

- All three pieces depend on one thing the government doesn't have - enough technical staff to verify anything itself. So that job gets handed to a certifier the company being checked picks and pays.
- Right at the moment a company nears a dangerous capability threshold, its own hand-picked certifier signs off, and nobody independent is watching.
- Liability doesn't save us here either since a company can't pay for a catastrophe, so the lawsuit threat never actually deters the worst outcomes.
- Net result: paperwork that looks like oversight, but isn't.

The fix:

- **Check the hardware, not the paperwork.** Build reporting into the chips themselves, so it doesn't depend on hiring more government auditors. (Short term)

- **Take away the conflict of interest.** Companies contribute talent to designing the certifications across the entire range of frontier labs, but the process is double blind - they collaborate to design the spec for everyone. Companies pay into a shared fund; certifiers get assigned by rotation, not chosen by the company they're checking.
- **Split the job by severity.** Let courts and certifiers handle everyday harms. Give a small, focused agency (like an AISI) the sole power to halt a training run - but only for specific, defined catastrophic thresholds, reauthorized by Congress on a set schedule, and triggered automatically if a certifier is later caught missing something.

Big Problems:

Who watches the watchers? This plan concentrates two things: the reporting system and the halt authority. Too much power in one place unless there's a way to separate the halt authority or build into the certification marketplace an independent organization with halt capability. The best I can do: keep that agency's power narrow and time-limited, it's members approved by both labs and congress (non-political joint appointees) and back it up with courts and Congress - knowing those checks are slower than the agency itself - so slow in fact as to be a commercial deterrent to model release (think human speed as the bottleneck - even worse actually - court speed!)

Highest cost: to respond fast to a catastrophe, you have to trust one body's judgment before anyone can double-check it. The fix isn't to spread all the power out evenly - a fully distributed system would be safer from abuse but too slow to act in a real emergency. Different risks need different structures: slow and spread out for everyday problems, fast and centralized for the worst-case ones.

As always, James is human and can make mistakes!